
Back to entropy

Evaluating language models as language models

David Afonso Valente · Bottlecap AI

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How good is this model, for what we do?

Choosing a base model today means reading a scorecard.

- A language model is trained to do one thing: **predict text**.
- So the direct question is: how well does it predict *our* text?
- Instead we read MMLU-Pro, HellaSwag, GSM8K.

Llama-3.2-1B vs Qwen2.5-1.5B

HellaSwag	2.1 points apart
Adapted BPB on news	3.4% apart
GSM8K	55.3 points apart

Two models that model fresh English almost identically well differ by nearly a factor of ten on one exam. Which number describes the model?

Two reasons the scorecard misleads

1. Benchmarks are public

- Written before the model existed, for other purposes.
- Vendors optimise for the ones they report.
- Items and subject matter can reach pre-training data.
- Stronger model, or saw more of the exam?
No reader can tell.

2. They score a model before it is adapted

- Teams adapt a base model to their data first.
- What matters is how good it *becomes*.
- **Gemma-4-12B**, of fifteen models: 9th on GSM8K, 8th on MMLU-Pro, 13th on raw loss ...
- ... and **the best fine-tuned system of all**.

Why loss is the right quantity

There is no universal ground truth for “better”, so pick the principled yardstick

- Every base model was trained to minimise cross-entropy on text. Held-out loss is **its own objective**, measured out of sample.
- Cross-entropy is a proper scoring rule. In expectation over the domain's text p :

$$H(p, q) = \underbrace{H(p)}_{\text{same for every model}} + \text{KL}(p \parallel q)$$

so a difference in loss *is* a difference in distance from the text's distribution.

- The one obstacle: each model counts in its own tokens. Divide by **bytes**.

Bits per byte is not one metric among many. It is the training objective, put on a common scale.

Taking that seriously: three requirements

Bytes

every tokenizer
on one scale

Fresh, decontaminated text

no candidate
has trained on it

Equal adaptation

the same procedure
for every candidate

Decontamination, three layers:

- **Time**: text dated after every model shipped (Reddit and Hacker News, most of arXiv).
- **Hygiene**: exact dedup, quality filter, MinHash near-dup removal, 80/10/10 split with no overlap.
- **Membership screens**: Min-K++, context sensitivity, perturbation; the union of anything any model flags is quarantined for all (news snapshot).

Equal adaptation: the procedure, and what we ran

The method

-hparams sweep, default

- Per-model Optuna search: TPE sampler over LR, batch, warm-up, weight decay, LoRA rank, α , dropout.
- Successive halving prunes weak trials from 50 steps.
- Scored on the *validation* split; test never touched.
- Same number of trials for every model, no time cap.
- Winner trained until validation BPB stops improving (-no-time-limit).

What these results use

-hparams manual

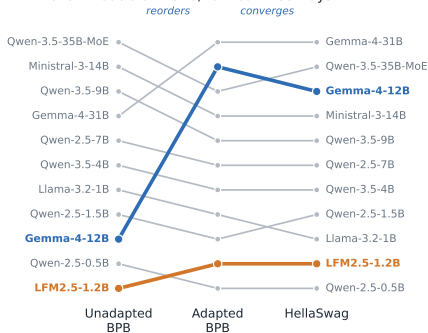
- A full search of 17 models up to 35B was beyond our compute.
- One hand-picked recipe: LoRA $r=16$, $\alpha=32$, lr 10^{-4} , batch 32, fixed 250 steps.
- One A100: **3.7 min** at 0.5B, 11.6 at 7.7B, 24.7 at 12B, 62.7 at 31B.
- Where our earlier search settled: rank 16 won for 8 of 11 models.

A time-capped search is a trap: small models finish more trials and get better tuned. In our first run the 1B models completed 17–18 trials, the two largest only 4.

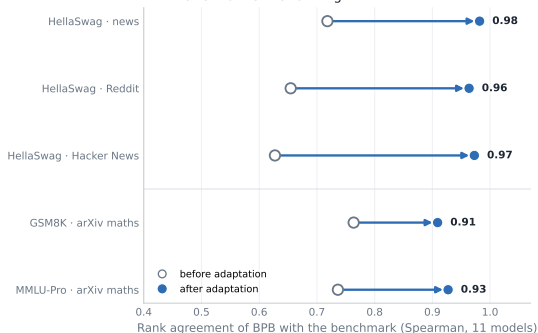
Adaptation reorders the models, and lands on the language benchmark

Left: eleven models ranked three ways on news. Right: what the adapted ordering tracks.

A. Eleven models on news, ranked three ways



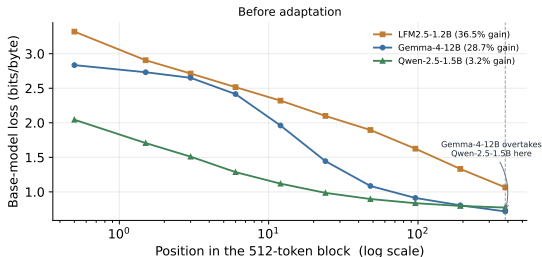
B. The text aims the ranking



Gemma-4-12B climbs from ninth to second; after adaptation the order is HellaSwag's (Spearman **0.96–0.98**). Adapted on maths instead, it swings to GSM8K and MMLU-Pro.

1. Adaptation reorders the models

Eleven models; unadapted vs adapted rankings agree only at $\rho = 0.61$ – 0.68



- Adaptation cuts loss by **3.6–5.4%** for every Qwen, Llama and Ministral model
...
- ... but by **17–36%** for the two Gemmas and LFM: the *only* models that climb.
- Gemma-4-12B: 9th → 2nd. Unadapted, it opens each block at 2.83 BPB vs a 1.5B model's 2.04, and passes it by the end.

Unadapted loss mixes *familiarity with this prose* into capability. A short equal adaptation removes the familiarity.

2. It converges on the language benchmark, and can be aimed

Rank correlation of BPB with each benchmark, before and after adaptation (eleven models)

	Before adaptation	After adaptation
HellaSwag, on news / Reddit / Hacker News	0.63–0.72	0.96–0.98
GSM8K, adapted on arXiv mathematics	0.76	0.91
MMLU-Pro, adapted on arXiv mathematics	0.74	0.93

- HellaSwag is the benchmark closest to language modelling, and adapted BPB lands on it.
- Change the text to mathematics and the ranking swings to the reasoning benchmarks.

3. It is stable where benchmarks are not

Adapted BPB

- Seed-to-seed SD is **0.0012**× the spread between models; 0 of 40 close pairs flip.
- Re-run weeks later on other hardware: within 0.0003.
- Adapted ranking across news / Reddit / Hacker News: $\rho = 0.955\text{--}0.991$.

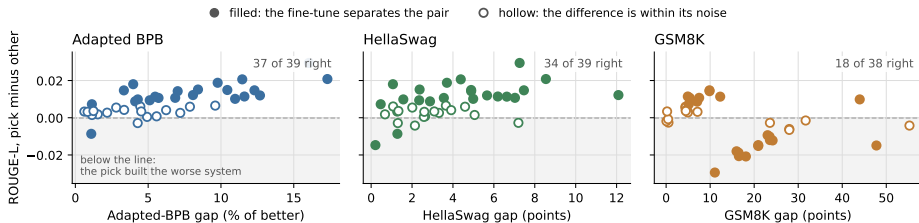
Generative benchmarks

- Same weights, new host (A100 → H100): GSM8K moves **−0.60 to +0.69** points.
- One shipped decoding setting: **9.86** GSM8K and **14.30** MMLU-Pro points for Gemma-4-12B.

A generative benchmark measures the model, the harness, the decoding and the hardware at once. Bits per byte on a fixed corpus measures the model.

4. Adapting to the domain predicts fine-tuning on the task

Fifteen models fine-tuned on news; 39 size-matched pairs



Adapted BPB, measured *before* any task existed, orders **0.949** of pairs as the fine-tune does, against HellaSwag 0.872, MMLU-Pro 0.513, GSM8K 0.474. **Every model within one rank** of where it finishes. Holds under a second, very different recipe (0.923).

Using it, and adding your model

- Candidates more than $\sim 2\times$ apart in size? The larger is usually better.
- Otherwise: recent domain text, decontaminated; adapt every finalist the same way; rank by adapted BPB.
- A gap under **2%** is a tie: let licence, latency or cost decide.
- The zero-shot number is a free screen, not the choice.

Public leaderboard, two tracks: [arXiv maths](#) (CC0 corpus in the repo; anyone can run it and submit) and [news](#) (maintainers run requested models). Submissions are checked against the track's corpus hash and recipe.

Evaluate language models as language models.

Questions?

Backup: per-model BPB, benchmark scores, contamination breakdown, longer-context transfer.

Backup: the adaptation results as a table

Lower BPB is better. Hand-picked recipe, 250 steps, news corpus.

Model	Architecture	Params	Zero-shot	Adapted	Reduction
Gemma-4-31B	Dense Transformer	31B	0.752	0.624	17.0%
Gemma-4-12B	Dense Transformer	12B	0.866	0.653	24.5%
Qwen-3.5-35B-MoE	Mixture of Experts	35B	0.707	0.678	4.1%
Ministral-3-14B	Dense Transformer	14B	0.718	0.679	5.4%
Qwen-3.5-9B	Dense Transformer	9B	0.740	0.708	4.4%
Qwen-2.5-7B	Dense Transformer	7B	0.758	0.717	5.4%
Qwen-3.5-4B	Dense Transformer	4B	0.782	0.749	4.2%
Llama-3.2-1B	Dense Transformer	1.2B	0.811	0.771	4.9%
Qwen-2.5-1.5B	Dense Transformer	1.5B	0.833	0.797	4.3%
LiquidAI-LFM2.5	Liquid Neural Net	1.2B	1.340	0.852	36.4%
Qwen-2.5-0.5B	Dense Transformer	0.5B	0.935	0.901	3.6%

Every value reproduces from the marker-masked fixed-protocol files in `results/domain_transfer/news_*.json`.

Backup: full benchmark scores

Eleven-model cohort, one host, one build

Model	Adapted BPB	MMLU-Pro	HellaSwag	GSM8K
Gemma-4-31B	0.624 (1)	57.1 (2)	87.9 (1)	85.0 (3)
Gemma-4-12B	0.653 (2)	44.8 (7)	84.0 (3)	64.8 (7)
Qwen-3.5-35B-MoE	0.678 (3)	61.4 (1)	85.6 (2)	86.6 (1)
Ministral-3-14B	0.679 (4)	52.8 (4)	82.9 (4)	80.8 (6)
Qwen-3.5-9B	0.708 (5)	55.6 (3)	81.6 (5)	85.8 (2)
Qwen-2.5-7B	0.717 (6)	47.2 (6)	80.3 (6)	81.3 (4)
Qwen-3.5-4B	0.749 (7)	52.4 (5)	77.2 (7)	81.0 (5)
Llama-3.2-1B	0.771 (8)	11.2 (11)	65.8 (9)	6.4 (11)
Qwen-2.5-1.5B	0.797 (9)	25.6 (8)	68.0 (8)	61.6 (8)
LiquidAI-LFM2.5	0.852 (10)	16.4 (9)	60.9 (10)	54.1 (9)
Qwen-2.5-0.5B	0.901 (11)	16.0 (10)	51.8 (11)	34.3 (10)

Percent, rank in parentheses; generated from the committed harness logs, the same table as the paper's. *lm-evaluation-harness* 0.4.12, bf16, greedy, native BOS; MMLU-Pro 5-shot CoT on a fixed 1,000-question subset, HellaSwag 10-shot length-normalised, GSM8K 5-shot flexible-extract.

Backup: contamination breakdown

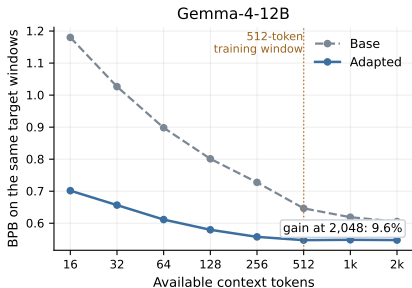
$N = 1000$ sampled sequences per model; 80 unique risk-flagged sequences removed

Model	Min-K++ flags	CoDeC flags	DCQ candidates	Preference	Risk flags
google-gemma-4-31B	12	6	20	100%	20
google-gemma-4-12B	11	6	11	100%	11
mistralai-Ministral-3-14B	12	8	13	100%	13
Qwen-Qwen3.5-35B-MoE	6	8	11	100%	11
Qwen-Qwen3.5-9B	12	6	17	100%	17
Qwen-Qwen2.5-7B	10	6	14	100%	14
Qwen-Qwen3.5-4B	9	6	12	100%	12
meta-llama-Llama-3.2-1B	17	0	28	100%	28
Qwen-Qwen2.5-1.5B	14	4	17	100%	17
LiquidAI-LFM2.5	12	8	23	96%	22
Qwen-Qwen2.5-0.5B	8	2	13	100%	13

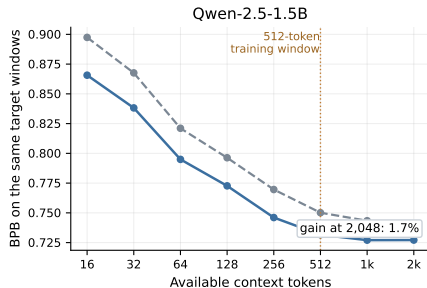
These are conservative quarantine signals on a preselected subset, not verified memorization rates. Exact overlap and model-likelihood signals must be interpreted separately.

Backup: 512-token adaptation transfers to longer contexts

Same held-out targets; only the available preceding context changes



Gemma base BPB keeps recovering after 512:
0.6465 → **0.6054** at 2,048 tokens.



Adapted Gemma stays at **0.547**; its advantage remains **9.6%** at four times the training context.

The adapter does not require the inference sequence to stop at 512 tokens.